

User Acceptance Testing (UAT) Template

Date	9-03-2026
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	

Project Overview

Project Name: AI-Driven Healthcare Anomaly Detection System

Project Description:

The AI-Driven Healthcare Anomaly Detection System is a real-time monitoring platform designed to detect abnormal patterns in patient vital signs and generate early health alerts. The system processes physiological data such as heart rate, oxygen saturation (SpO₂), temperature, and blood pressure using machine learning models like Isolation Forest and Autoencoder Neural Networks. Detected anomalies are stored in a PostgreSQL database and visualized through a Flask-based dashboard, while critical alerts trigger automated notifications to healthcare administrators.

Project Version: Version 1.0

Testing Period: 5th March 2026 – 8th March 2026

Testing Scope

Features and Functionalities to be Tested

- Patient vital data ingestion
- Data preprocessing and cleaning
- Machine learning anomaly detection models
- Dashboard visualization
- Database storage and retrieval
- Email alert notification system
- API communication between system components

User Stories / Requirements to be Tested

- Data collection and loading functionality
- Data preparation and preprocessing
- Data visualization through charts and dashboard
- Anomaly detection functionality
- System **performance under multiple requests**

Testing Environment:

URL/Location: https://github.com/Rutujjs/GEN_AI_SMARTBRIDGE

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
TC-001	Data Input Verification	Step 1: Enter patient vital parameters Step 2: Submit data	System should accept valid inputs	System accepted inputs successfully	Verified
TC-002	Data Processing	Step 1: Run data preprocessing module Step 2: Handle missing values	System should clean and process data	Data processed correctly	Verified
TC-003	Anomaly Detection Model	Step 1: Run anomaly detection algorithm Step 2: Generate anomaly score	System should detect abnormal patterns	Anomaly detection executed correctly	Verified
TC-004	Dashboard Visualization	Step 1: Open dashboard Step 2: View charts and anomaly status	Dashboard should display results	Dashboard displayed results successfully	Verified
TC-005	Database Storage	Step 1: Run system Step 2: Check PostgreSQL database	Data should be stored correctly	Data stored successfully	Verified

Bug Tracking:

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	Minor delay during dashboard loading when processing large data	Run dashboard with large dataset	Low	Resolved	Performance improved after optimization

Sign-off:

Tester Name: [Rutuja Pote]

Date: [5th march-8th march]

Signature: [Rutuja Pote]